

SHREETI SHRESTHA

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[LinkedIn](#) | [Github](#) | [Personal Website](#)

EDUCATION

Northeastern University, Boston, MA

Jan 2025 - Present

Master of Science in Computer Science | GPA: 3.9/4.0

Anticipated Graduation: May 2027

Coursework: Machine Learning, Advanced Machine Learning, Actionable Interpretability, Research in Human-Centered NLP, Foundations of Responsible AI, Scalable Distributed Systems, Algorithms, Intro to Deep Learning (MIT 6.S. 191)

Ramapo College of New Jersey, Mahwah, NJ

Aug 2018 - Dec 2021

Bachelor of Science in Computer Science | GPA: 3.85/4.0

Coursework: Machine Learning, Statistics, Linear Algebra, Discrete Structures, Operating Systems, Data Structures

TECHNICAL KNOWLEDGE

ML/AI Frameworks :PyTorch, Hugging Face, scikit-learn, TensorFlow, SAELens, NDIF, Neuronpedia, Transformers

Domains :NLP, Deep Learning, Time Series Analysis, Anomaly Detection, Statistical Forecasting, LLM

Languages :Python, C++, Java, JavaScript, C#, Go

Cloud & Databases :AWS (EC2, S3, Lambda), Docker, PostgreSQL, MongoDB, MySQL , Git

RESEARCH EXPERIENCE/PROJECTS

Graduate Research Assistant | UbiWell Lab

Aug 2025 – Present

Northeastern University, Boston, MA

- Developing real-time **anomaly detection** on longitudinal smartphone sensing data across 80+ participants, training and evaluating isolation forests and **LSTM sequence models** for behavioral pattern representation learning
- Built **AI dashboards** to monitor performance, representation drift, and trigger quality for mental-health interventions
- Co-authored and presented a research poster “Real-Time Anomaly-driven EMAs to Capture Varying Mental and Behavioral States among Adolescents” at Digital Health Summit, Dartmouth University (Dec 2025) | [Link](#)

Adaptive Tutoring with LLMs: Comparative Analysis Against Traditional Methods | [Link](#)

April 2025

Research in Human-Centered NLP, Course Project

- Fine-tuned a **transformer-based LLM** for adaptive LSAT tutoring, using few-shot learning and prompt engineering
- Conducted **A/B testing** and evaluation using paired t-tests and Cohen’s d to quantify and compare learning gains

Evaluating Model Outputs with NLP Metrics and Human Feedback for Medical Summaries | [Link](#) February 2025

- Evaluated 150 summaries generated by a fine-tuned LLM using NLP metrics such as ROUGE, BLEU, BERTScore
- Analyzed failure modes and hallucination patterns through human feedback to inform prompt refinement strategies

Experimental Design for Evaluating Source Attribution in Health Chatbots | [Link](#)

March 2025

- Designed a split-plot study to evaluate impact of source citations on chatbot trust and usability across users
- Defined metrics and proposed quantitative (t-test, Wilcoxon) and qualitative thematic analysis of debrief responses

An Ontology for Psychological Ownership to Predict Organizational Ambidexterity | [Link](#)

March 2021

Ramapo College of NJ; Presented at Society of Advanced Management

- Modeled psychological and behavioral markers to predict organizational ambidexterity

WORK EXPERIENCE

Software Engineer

Jan 2022 - Dec 2024

Trimble Maps, Princeton, NJ (Industry: Routing and Navigation for commercial delivery vehicles)

- Improved **ETA prediction** by 20% by deploying large-scale ML systems, including feature engineering for spatio-temporal dependencies, training and evaluating **clustering and random forest** models on 150k+ daily routes
- Engineered high-throughput **AWS ETL pipelines** processing 150k+ daily routes with **low-latency requirements**, enabling real-time analytics and rapid model iteration
- Led **end-to-end development** of Voyager, a tool for **real-time monitoring** and evaluation of routing algorithms
- Notable Achievements: 2022 MVP of the Year, 2023 Top 10 Hackathon Submissions

Data Engineer Intern

May 2021 - Dec 2021

IQVIA, Parsippany, NJ (Industry: Healthcare Analytics)

- Built statistical forecasting models using Python to predict market trends with 90% confidence intervals, incorporating time series analysis and **regression techniques** for large-scale pharmaceutical prescription data
- Boosted operational efficiency by 15% by implementing real-time **anomaly detection** and operational dashboards